

Daanish Khan

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EDUCATION

Columbia University

MS in Computer Science

New York, NY

Aug 2025 - Dec 2026

Carnegie Mellon University

BS in Computer Science, Minor in Tech Entrepreneurship

Doha, Qatar

Aug 2021 - May 2025

Relevant Course Work: Networks, Embedded Systems, Algorithms, Vector Calculus, Linear Algebra.

ACADEMIC PROJECT & EXPERIENCE

Autonomous Robot SLAM & Navigation System

- Devised and implemented a closed-loop SLAM pipeline in Python for autonomous exploration of indoor environments leveraging an RGB-D equipped mobile robot.
- Implemented multi-view 3D point cloud reconstruction by applying rigid transformations to camera-space depth observations into a global world frame, utilizing NumPy for matrix operations and point cloud fusion.
- Maintained a 2D occupancy grid with 3 discrete states (FREE / OCCUPIED / UNKNOWN) via ray-casting-based updates, and planned collision-free paths using A* and probabilistic roadmaps for autonomous navigation.

Real-Time N-Body Gravity Simulation & Rendering Engine

- Developed a real-time N-body gravitational simulation in C++ modeling Newtonian dynamics among multiple celestial bodies, performing pairwise force accumulation and timestep-based numerical integration.
- Constructed a complete OpenGL rendering pipeline with GLSL vertex and fragment shaders, including Lambertian shading, per-vertex normal computation, and procedural sphere mesh generation.
- Created a full 3D camera system deploying 3 matrices (model, view, projection) to map world-space geometry to screen space, and analyzed performance and numerical stability as the number of simulated bodies scaled.

Peer-to-Peer File Transfer System (C, UDP)

- Engineered a serverless peer-to-peer file transfer system in C, where peers discover chunks by broadcasting SHA-1 hashes (WHOHAS / IHAVE / GET) and exchange 512KB chunks over a custom 16-byte packet format built directly on UDP.
- Implemented reliable delivery on top of an unreliable transport using sequence and acknowledgement numbers, an 8-packet sliding window with per-slot retransmit timers, and duplicate and out-of-order detection, validated against a simulated network applying per-link bandwidth, delay and packet loss.

EXPERIENCE

Matar, Hamad International Airport, Baggage Handling System

Doha, Qatar

Tech Consultant Intern

Jun 2025 - Aug 2026

- Developed a machine learning roadmap for HIA's baggage handling system to enhance bag tag reading accuracy and cut baggage mishandling rates by up to 50%.
- Reviewed Beumer Group's BHS documentation, in collaboration with 6 technical heads, to identify system bottlenecks and technical gaps, informing AI integration strategy.
- Presented AI roadmap to 3 of HIA leadership, showcasing its potential to modernize a legacy baggage system into a data-driven operation.

Brisa - Wearable Body Cooling Device

Doha, Qatar

Co-founder

Jan 2025 - Jul 2026

- Designed and prototyped a wearable cooling device leveraging Peltier thermoelectric modules, reducing skin surface temperature by 10°C to mitigate heat stress for Middle Eastern outdoor workers.
- Accelerated product readiness for field testing by engineering custom PCBs in KiCad and coding Arduino microcontrollers, driving technology development through iterative prototyping and user feedback.
- Earned recognition through multiple awards, including QBA's 'Most Valuable Product' and 'Investors' Choice Award,' validating Brisa's innovation and market potential.

Open Stack, Object Storage-Swift

Online, Global

Intern

May 2024 - Jul 2024

- Implemented byte-based quota in OpenStack Swift using Python, enabling administrators to allow storage limits by data size in addition to object count.
- Enhanced reliability of account reaper daemon by adding a recon dump "heartbeat" every 60 seconds, giving operators visibility into cleanup tasks and ensuring consistent account deletion.
- Tested new quota and reaper features in Oracle VM VirtualBox using Swift APIs, verifying accurate byte-limit enforcement and reliable heartbeat monitoring.
- Collaborated code via GerritHub peer review, with a team of 5, iterating on community feedback and aligning with OpenStack's open-source development standards.

SKILLS

Languages: C/C++/C#, Python, OpenGL, SML, Matlab, SQL, Dart, Java, Haskell, Bash.

Other Tools: KiCad, Arduino, Microcontroller application, Powershell.